WEEKLY PROGRESS REPORT

OCTOBER 2026 • WEEK 4 (22-28 OCTOBER 2026)

Project ID: MKCE/R&I/Consultancy/2026/001

AI-Based Egg Tray Inventory Management System (EG-Counter)

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1. Summary of the Week

All recorded work this week happened on 26 October, when the project moved to a new laptop. The team re-measured where everything stood, corrected some earlier claims that had been too strong, and wrote a step-by-step plan. With the user's approval they then put the new cloud server version (0.4.12) on the test server, with a lock so that only approved phones can send counts, and built the matching Android app with a new signing key. The new app was not installed on the phone this week, and no count has yet been checked against a physical recount of trays, so accuracy is still unknown.

Phase: Rolling out version 0.4.12 to the test (staging) server.

2. Work Completed This Week

2.1 Project re-checked on a new laptop

The code was freshly copied (main at 1800db9) onto a new laptop. The test server was found still on 47cb2378 (0.4.11) and the live (production) server on 4c1480e6. Files that git does not store did not come across, among them the test server's admin token, tool folders and the old APKs. Every automated test passed, and the saved results of earlier experiments came out byte-identical when re-run.

2.2 Earlier claims corrected

The earlier figure "393 vs 233 (-41%)" compared trays that were assumed, not seen (assistant 240 vs app 144). Only 153 trays were seen, so it does not measure how many layers the detector misses. A "164" reading was visual, with 60 trays assumed. Neither stored manual count can serve as the true count: MECH-DONE equals the app's own 399 because the form was pre-filled, and WH-99 was typed in by the assistant.

2.3 New cloud server version on the test server

With the user's explicit approval, an admin secret was set and version 0.4.12 of the cloud server (Cloudflare Worker) went onto the test server as version ef7c1dd3 (100%, 2026-10-26T15:57Z). It runs in "observe" mode: it checks phone identity, but phones still on the old app keep working. Before the change, the 2 stored manual-count records (25 stack rows) were saved. Checks used no paid image analysis, and requests without credentials were refused (401).

2.4 New signing key and signed app file built

The old app signing key exists only on the old (DELL) machine, so the user chose a new release key (RSA 4096, valid to 2054-03-13), kept outside the repository. APK 0.4.12 (versionCode 23, 62,542,361 bytes) was built in 824 s and its signature was verified. Neither the admin token nor the key password is inside the APK. It was not installed because no phone was connected.

2.5 Next plan written (tracks A-D)

A: engineering clean-ups the assistant can do alone (explicit deploy commands, scorer fixes, a register of held-back test photos, a field-test protocol fixed before any data). B: the 0.4.12 rollout in 7 steps. C: a field round on at least 3 new blocks of stacks, with independent per-stack counts. D: one retraining of the detector on far and cross-stacked faces.

3. Measured Results

Measurement	Value	What it shows
Automated software tests on the new laptop	Worker 111/111; app analyze clean, tests 167/167; Python backend 148 passed (4/4 runs, no OpenCV flake), ml 3, scripts 75	The server, app and analysis code pass their built-in automated checks on the new machine. These are software tests only. They do not measure how accurately the app counts trays.
Re-run of earlier experiment results	eval-20261015, walk-fill and stack-faces results all reproduce byte-identically; walk-fill results.json hash 77f4b6d0...	Re-running the earlier experiments on the new laptop gave exactly the same output files, so the old results can be repeated. (The hash was also re-checked for this report.) It adds no new accuracy evidence.
Access checks on the test server after the 0.4.12 deploy	No credentials: GET /v1/devices/me and /v1/scans/<id>/result return 401 device_credentials_required; GET /v1/manual-counts and /v1/admin/devices return 401 admin_auth_failed; with the admin token /v1/admin/devices returns 200 {devices: []}	The test server now refuses requests that lack a phone or admin credential, and the admin token works. The checks used no paid image analysis. The live (production) server was not changed.
Signed app file (APK 0.4.12)	62,542,361 bytes; versionName 0.4.12, versionCode 23; apksigner verifies (v2), 1 signer; `flutter build apk --release` took 824 s	The app file was built and its signature checks out with the new key. It was not installed on any phone this week, so it has not been tested on the phone.
Physical (hand) recounts available as a true reference	0 physical counts	No app count has yet been compared with a real physical recount of the trays. The two stored manual counts were judged unusable as truth, so the app's accuracy is still not measured.

None of these figures is a physical recount of trays: the references are handwritten labels, counts made by eye from photos, or software checks, as each row says.

4. Challenges Faced

• Moving to a new laptop lost files that git does not keep, including the test server's admin token, tool folders and the old APKs. A fresh copy of the server code would not build until a setup step (`npx wrangler types`) was run. The default deploy command (`npm run deploy`) was found to point at the live production server.

• The old app signing key is only on the old DELL machine, so a new key was made. An app signed with the new key cannot update the installed 0.4.11, so the user has to export counts as CSV and uninstall, which deletes the on-phone history. Until the user backs it up, the only copy of the new key is on this single-drive laptop.

• There is still no reliable reference count. Both stored manual counts are unusable as truth (one came from a pre-filled form, the other was entered by the assistant), and there are 0 physical counts. Accuracy therefore cannot be stated yet.

• The build setup was slow and needed manual steps. At first only 0.7 to 8.5 GB of disk was free (later 42 GB). The Gradle tool download timed out twice. The automatic permission check refused to set the server secret until the user approved it. No phone was connected, so the install had to wait.

5. Plan for the Next Week

• The user copies all counts as CSV in the old app (0.4.11), then uninstalls it.

• Connect the Redmi phone with USB debugging and install APK 0.4.12 (`adb install egg-tray-counter-0.4.12-block-staging.apk`).

• In the app's Settings, set the test (staging) server address, then approve the phone by its full id (`scripts/devices.py approve`).

• Run NETWORK TEST (watching `adb logcat | findstr EGGNET`), then one scan with RECORD MANUAL COUNT.

• Switch the test server from observe to enforce mode, so only approved phones are accepted.

• After that: a field round on at least 3 new blocks, photographed STRAIGHT/LEFT/RIGHT at about 1.5/3/5 m, with independent per-stack filled/empty counts (ideally two counters), to get the first held-out accuracy number. Then one detector retraining on far and cross-stacked faces.

6. Status at the End of the Week

App: APK 0.4.12 (versionCode 23), built and signed with the new release key but not installed; the phone still runs 0.4.11. Cloud server (Worker): test server (staging) runs 0.4.12 in observe mode as version ef7c1dd3 (deployed 2026-10-26T15:57Z); the live server (production) is unchanged at 4c1480e6 (2026-09-23).

Evidence: PROGRESS.md (entry "2026-10-26 - new machine: state re-measured, next plan", lines 508-668); context.md (2026-10-26 notes at the top, committed in 3989f95); git commit 3989f95 (2026-10-29: records this week's 26 Oct work; no commits are dated 22-28 Oct); mobile/android/app/build.gradle.kts (release builds sign with key.properties, changed in 3989f95); docs/SCAN_JOBS_AND_DEVICES.md (meaning of observe vs enforce mode); reports/experiment-walk-fill-20261015/results.json (SHA-256 77f4b6d0..., re-checked); reports/eval-20261015/ and reports/experiment-stack-faces-20261015/ (results said to reproduce; these folders contain no images).